Burns — Children: Fluids, TBSA by Age & Nutrition

children with burns · casualty / ward

4 mL × kg × % + maint

FLUIDS (FROM TIME OF BURN)

Resuscitation 24 h ✚: 4 mL × kg × %TBSA (NaCl 0.9% or Ringer's lactate); ½ in first 8 h

PLUS maintenance ✚: ≤1 y: 120 mL/kg/24 h · >1 y: 100/50/20 mL/kg rule (dextrose-saline)

Target urine output ✚: 1–2 mL/kg/h; if inadequate ↑ fluids by 50% for the next hour

Day 2 ✚: NaCl 0.9% 1.5 mL × kg × % + maintenance

Energy ✚: 250 kJ/kg + 150 kJ × %TBSA per day

Protein ✚: 3 g/kg + 1 g × %TBSA per day (max %TBSA 50)

TBSA % BY AGE (YEARS) — TORSO FRONT 18 + BACK 18, EACH ARM 9 AT ALL AGES ✚

Age	Head+neck	Each leg	Age	Head+neck	Each leg
<1	18	14	5–<6	13	16.5
1–<2	17	14.5	6–<7	12	17
2–<3	16	15	7–<8	11	17.5
3–<4	15	15.5	8–<9	10	18
4–<5	14	16	≥9	9	18

WORKED EXAMPLE 24 H (RESUS + MAINTENANCE) ML

kg \ %	10%	15%	20%	30%
10	400 + 1000	600 + 1000	800 + 1000	1200 + 1000
15	600 + 1250	900 + 1250	1200 + 1250	1800 + 1250
20	800 + 1500	1200 + 1500	1600 + 1500	2400 + 1500
24	960 + 1580	1440 + 1580	1920 + 1580	2880 + 1580
30	1200 + 1700	1800 + 1700	2400 + 1700	3600 + 1700

MAJOR BURNS — ADMIT / REFER

- Partial thickness >10% or full thickness >3% ✚
- Head, face, hands, feet, perineum; circumferential; inhalation; electrical ✚
- Neonates; comorbid injuries ✚
- Minor = partial thickness <10% in a child >1 year ✚

SA STG — MANAGEMENT

- Soak in cold water for not more than 20 min; oxygen in all burns >10% or possible CO poisoning ✚
- Child's palm = 1% TBSA; adult percentages from age 9 (plus 1% perineum) ✚
- NG tube on free drainage in all burns >10%; start enteral feeds within 6 h ✚
- Stop resuscitation fluids if UO >1–2 mL/kg/h or base excess better than −4 ✚
- Hb <7 g/dL: packed red cells 10 mL/kg over 3 h; symptomatic hypoalbuminaemia: albumin 20% 2 g/kg/day ✚